

Notes: Allcott, Braghieri, Eichmeyer & Gentzkow (AER, 2020)

The Welfare Effects of Social Media

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1 Big picture

- **Question.** What are the welfare effects of Facebook for individual users and for political/social outcomes?
- **Design.** A randomized encouragement experiment paid Facebook users to deactivate accounts for the four weeks before the 2018 US midterm election.
- **Main results.** Deactivation reallocates time offline, reduces news knowledge, reduces some forms of political polarization, increases subjective well-being, and lowers future Facebook use and valuation.
- **Overall message.** Facebook provides large consumer value but also creates costs: lower well-being, more polarization, and possible overuse relative to users' own long-run preferences.

2 Incremental contribution of this paper

- **Scale and power.** Relative to prior social-media deactivation experiments, this paper uses a much larger sample and a longer four-week deactivation window.
- **Comprehensive welfare scope.** It measures time use, news knowledge, political engagement, polarization, subjective well-being, future demand, opinions, and WTA in one unified design.
- **Election timing.** The experiment is timed before the 2018 US midterms, giving political-information and polarization outcomes substantive relevance.
- **Randomized encouragement plus verified compliance.** The authors verify deactivation and use randomized treatment assignment as an instrument for actual deactivation.
- **Pre-analysis plan.** Outcomes, indices, and moderators were specified before endline, reducing researcher degrees of freedom.
- **Demand and welfare angle.** The repeated WTA design allows the paper to connect treatment effects to consumer surplus, learning, habit formation, and projection bias.

3 Data, experimental design, and measurement

3.1 Recruitment, sample, and assignment

Participants were recruited through Facebook ads. The impact-evaluation sample consists of users willing to accept less than \$102 to deactivate and then randomized to a \$102 treatment offer or \$0 control offer. The treatment window covered the four weeks before the 2018 midterm election.

Interpretation. The sample is selected into people willing to consider deactivation, but the randomized offer generates clean treatment variation within that group.

3.2 Outcome families

The paper studies substitute time uses, social interaction, substitute news sources, news knowledge, political engagement, political polarization, subjective well-being, post-experiment use, and opinions about Facebook. Outcomes are combined into Anderson-style inverse-covariance weighted indices.

Interpretation. The index structure reduces multiple-testing concerns and makes families of related outcomes easier to interpret.

3.3 Compliance and measurement

Compliance is measured as the share of deactivation checks on which a participant was observed to be deactivated. Treatment compliance was high: treatment participants were deactivated on about 90% of checks, versus 2% for controls.

Interpretation. High compliance and low attrition make the randomized encouragement design unusually strong for studying platform use.

4 Models and empirical specifications

4.1 Main LATE specification

$$Y_i = \tau D_i + \rho Y_i^b + \nu_s + \varepsilon_i,$$

where D_i is instrumented with treatment assignment T_i .

Interpretation. The coefficient τ is the local average treatment effect for users induced to deactivate by the \$102 payment.

4.2 Heterogeneous treatment effects

$$Y_i = \tau D_i + \alpha_k D_i X_i^k + \zeta X_i^k + \rho Y_i^b + \nu_s + \varepsilon_i,$$

where D_i and $D_i X_i^k$ are instrumented with T_i and $T_i X_i^k$.

Interpretation. The interaction coefficient α_k measures whether the treatment effect differs across baseline characteristics.

4.3 WTA updating

$$\Delta w_{2,i} = \gamma D_i + \rho w_{1,i} + \nu_s + \varepsilon_i,$$

where D_i is instrumented with T_i .

Interpretation. A negative γ means that experiencing deactivation lowered future valuation of Facebook.

5 Identification and empirical design

- Randomized assignment to the \$102 deactivation offer creates exogenous variation in deactivation.
- The first stage is strong: treatment users deactivate on roughly 90% of checks.
- The main exclusion restriction is that the offer affects outcomes through deactivation, not through experimenter demand or payment effects.
- The pre-analysis plan limits specification search.
- External validity is limited by sample selection, four-week duration, and possible general-equilibrium effects of mass deactivation.

6 Tables and figures

6.1 Figure 1: Experimental Design + Table 1: Sample Sizes

Read: Figure 1 lays out recruitment, baseline, treatment/control deactivation, endline, and post-endline measurement. Table 1 shows the funnel from 1.89 million users shown ads to 1,661 impact-evaluation subjects.

Economic message. The design combines incentive-compatible valuation, randomized encouragement, verified compliance, and repeated outcome measurement.

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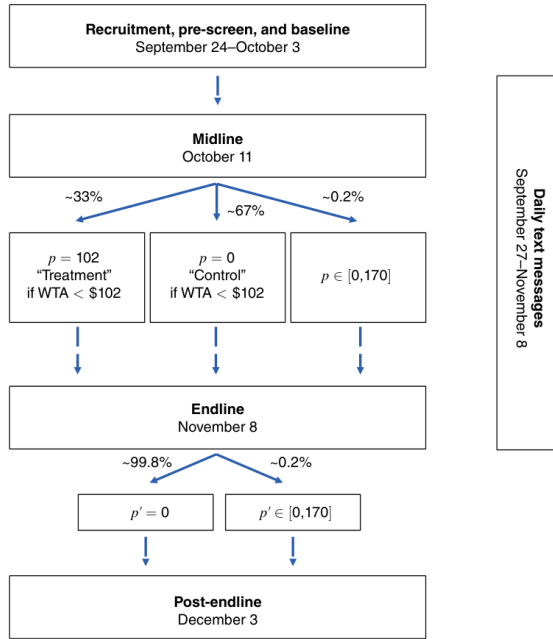


FIGURE 1. EXPERIMENTAL DESIGN

(a) Figure 1: Experimental Design

TABLE 1—SAMPLE SIZES	
Phase	Sample size
Recruitment and baseline	$N = 1,892,191$ were shown ads
	$N = 32,201$ clicked on ads
	$N = 22,324$ completed pre-screen survey
	$N = 20,959$ were from United States and born between 1900 and 2000
	$N = 17,335$ had $15 < \text{daily Facebook minutes} \leq 600$
	$N = 7,455$ consented to participate
	$N = 3,910$ finished baseline
Midline	$N = 2,897$ had valid baseline and were randomized, of which:
	$N = 2,897$ began midline
	$N = 2,743$ received a price offer, of which:
Endline	$N = 1,661$ were in impact evaluation sample
	$N = 2,710$ began endline
	$N = 2,684$ finished endline, of which:

(b) Table 1: Sample Sizes

6.2 Table 2: Sample Demographics + Table 3: Survey Response and Treatment Compliance Rates

Read: Table 2 shows external validity and sample selection. Table 3 shows strong compliance and low attrition.

Economic message. The sample is selected, but treatment implementation and follow-up are unusually strong.

	Impact evaluation sample (1)	Facebook users (2)	US population (3)
Income under \$50,000	0.40	0.41	0.42
College	0.51	0.33	0.29
Male	0.43	0.44	0.49
White	0.68	0.73	0.74
Age under 30	0.52	0.26	0.21
Republican	0.13	0.26	0.26
Democrat	0.42		0.20
Facebook minutes	74.52	45.00	

Notes: Column 1 presents average demographics for the impact evaluation sample: participants who were willing to accept less than \$102 to deactivate Facebook for the four weeks after midline and were offered $p = \$102$ or $p = \$0$ to do so. Column 2 presents our estimate of average demographics of American adults with a Facebook account. The top five numbers in column 2 are inferred from a Pew Research Center (2018f) survey of social media use by demographic group. The bottom number in column 2 (the average of 45 minutes of Facebook use per day) is approximated on those basis of sources such as Facebook (2016) and Molla and Wagner (2018). Column 3 presents average demographics of American adults. The top five numbers are from the 2017 American Community Survey (US Census Bureau 2017), and the Republican and Democrat shares are from the 2016 American National Election Study (American National Election Studies 2016).

(a) Table 2: Sample Demographics

Variable	Treatment mean/SD (1)	Control mean/SD (2)	<i>t</i> -test <i>p</i> -value (1) – (2)
Completed endline survey	0.99 (0.11)	0.98 (0.12)	0.54
Share of text messages completed	0.92 (0.20)	0.93 (0.18)	0.45
Completed post-endline survey	0.95 (0.23)	0.92 (0.26)	0.07
Share days deactivated	0.90 (0.29)	0.02 (0.13)	0.00
Observations	580	1,081	

Notes: Columns 1 and 2 present survey response and treatment compliance rates for the Treatment and Control groups in the impact evaluation sample: participants who were willing to accept less than \$102 to deactivate Facebook for the four weeks after midline and were offered $p = \$102$ or $p = \$0$ to do so. Column 3 presents *p*-values of tests of differences in response rates between the two groups.

(b) Table 3: Survey Response and Treatment Compliance

6.3 Figure 2: Substitutes for Facebook + Figure 3: Effects on News and Political Outcomes

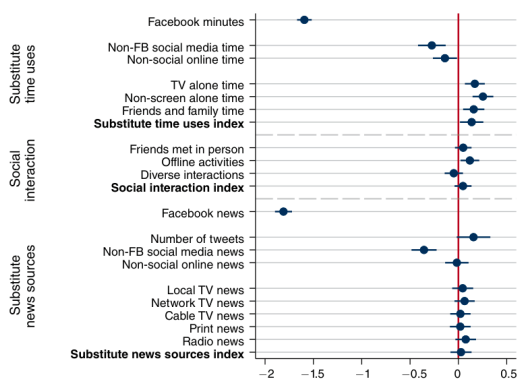
Read: Figure 2 shows time reallocation away from Facebook and other online activity toward offline uses. Figure 3 shows reduced news knowledge and reduced polarization by some measures.

Economic message. Facebook is an important information channel, but also appears to contribute to polarizing exposure.

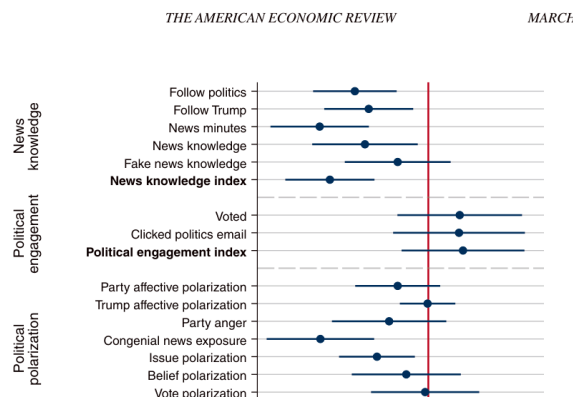
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(a) Figure 2: Substitutes for Facebook



(b) Figure 3: News and Political Outcomes

6.4 Figure 4: Issue Opinions by Party at Endline + Figure 5: Effects on Subjective Well-Being

Read: Figure 4 visualizes reduced issue polarization. Figure 5 shows positive effects on subjective well-being.

Economic message. Deactivation makes users less informed but less polarized and somewhat happier.

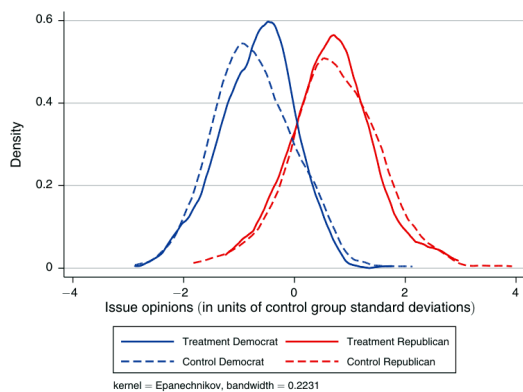
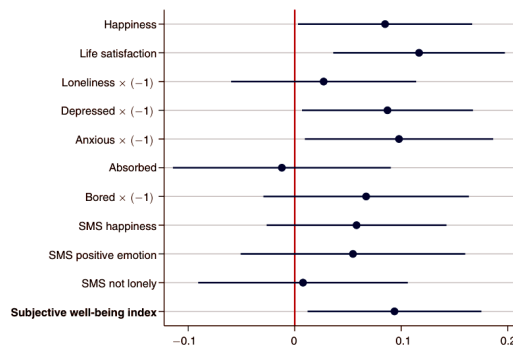


FIGURE 4. ISSUE OPINIONS BY PARTY AT ENDLINE

Notes: This figure presents kernel density plots of issue opinions for Democrats and Republicans in Treatment and Control at endline. Issue opinions are attitudes about nine current political issues on a scale from -5 to $+5$.

(a) Figure 4: Issue Opinions by Party at Endline

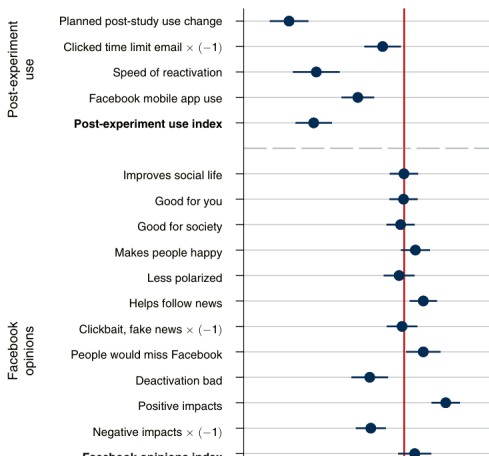


(b) Figure 5: Subjective Well-Being

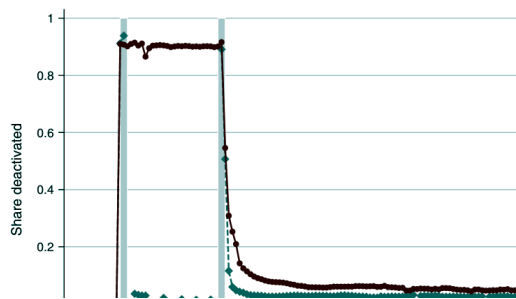
6.5 Figure 6: Post-Experiment Facebook Use and Opinions + Figure 7: Probability of Being Deactivated

Read: Figure 6 shows large persistent reductions in future Facebook use. Figure 7 confirms strong compliance during the experiment.

Economic message. Deactivation changes not only in-experiment outcomes but also users' future demand.



(a) Figure 6: Post-Experiment Use and Opinions

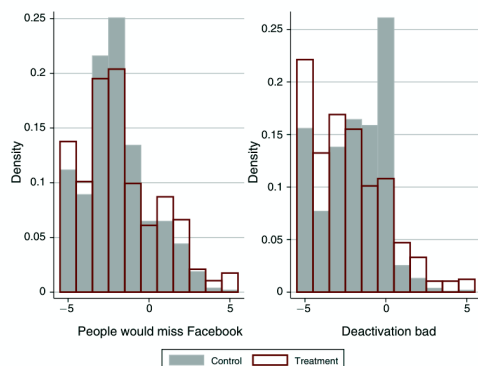


(b) Figure 7: Probability of Being Deactivated

6.6 Figure 8: Key Opinions about Facebook + Table 4: Most Common Descriptions of Facebook Use Changes

Read: Figure 8 shows many treated users thought deactivation was good for them. Table 4 shows qualitative language about how users changed behavior.

Economic message. The qualitative responses help explain the mechanisms behind changed use, well-being, and WTA.



(a) Figure 8: Key Opinions about Facebook

TABLE 4—MOST COMMON DESCRIPTIONS OF FACEBOOK USE CHANGES

Phrases used more often by Treatment			Phrases used more often by Control		
Phrase	% Treatment	% Control	Phrase	% Treatment	% Control
Not use facebook anymore	0.90	0	Ha not chang	6.63	16.76
Not spend much time	1.08	0.36	Not chang sinc particip	0	0.99
Spend less time facebook	0.90	0.27	Ha not chang sinc	0.18	1.53
Have not use facebook	0.72	0.18	Chang sinc particip studi	0	0.81
Not use facebook much	0.72	0.18	Way use facebook ha	0.18	1.35
Spend lot less time	0.72	0.27	Usag ha not chang	0	0.72
Use much less	2.87	0.63	Chang way use facebook	0.18	1.26
Definit use facebook	0.54	0.18	Not chang	7.17	18.65
Use facebook lot less	0.54	0.18	Awar much time spend	0	0.63
Use facebook much less	0.54	0.18	Ha not	8.24	19.64
Not use facebook	3.05	1.17	Not much ha chang	0	0.54
Use littl bit less	0.54	0.27	Way use facebook	0.54	2.70
Have not use	1.25	0.18	Not think chang much	0	0.45
Ha not chang use	0.72	0.45	Not chang much use	0	0.45
Use facebook anymore	0.90	0.09	Use facebook slightli less	0	0.45
Think use less	1.61	0.45	More awar much time	0.18	0.99
No ha not chang	0.54	0.36	Chang sinc particip	0	1.08
Use news app	0.72	0.09	Much time spend	0.18	1.53
Still have not	0.90	0.18	Facebook ha not chang	0.72	2.07
Much less	4.84	1.17	Use slightli less	0	0.90

Notes: The post-endline survey included the following question with an open response text box: "How has the way you use Facebook changed, if at all, since participating in this study?" For all responses, we stemmed words, filtered out stop words, then constructed all phrases of length $l = \{1, 2, 3, 4\}$ words. For each phrase p of length l , we calculated the number of occurrences of that phrase in Treatment and Control group responses (f_{pt} and f_{pc}) and the number of occurrences of length- l phrases that are *not* phrase p in Treatment and Control responses (f_{-pt} and f_{-pc}). We then constructed Pearson's χ^2 -statistic:

$$\chi^2 = \frac{(f_{pt}f_{-pc} - f_{pc}f_{-pt})^2}{(f_{pt} + f_{pc})(f_{pt} + f_{-pt})(f_{pc} + f_{-pc})(f_{-pt} + f_{-pc})}$$

This table presents the 20 phrases with the highest χ^2 that were most commonly written by the Treatment and Control groups. The % Treatment and % Control columns present the share of people in the respective group whose responses included each phrase.

(b) Table 4: Descriptions of Use Changes

6.7 Figure 9: Heterogeneous Treatment Effects

Read: Figure 9 tests pre-specified heterogeneity by heavy use, news use, active/passive use, and peak/off-peak timing.

Economic message. The strongest intuitive pattern is larger news effects for people who used Facebook for news; other moderators are weaker.

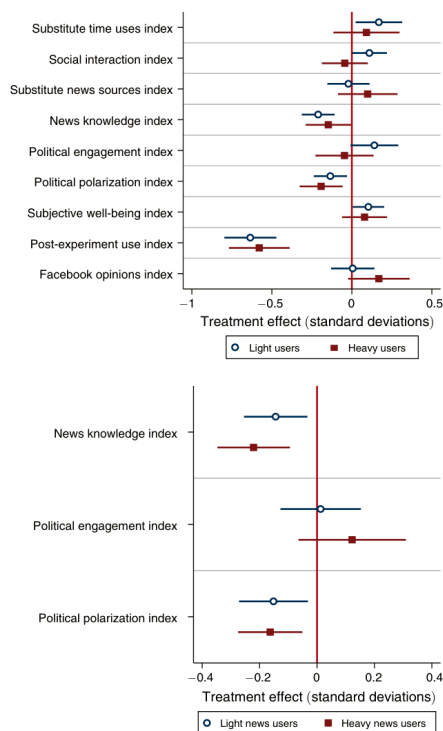


FIGURE 9. HETEROGENEOUS TREATMENT EFFECTS

(Continued)

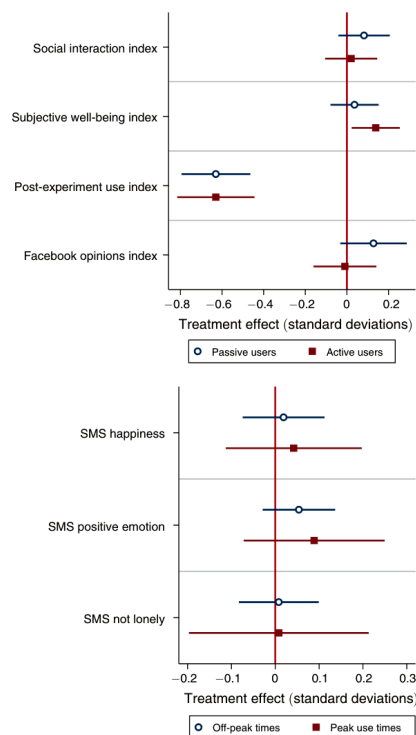


FIGURE 9. HETEROGENEOUS TREATMENT EFFECTS (CONTINUED)

Notes: This figure presents local average treatment effects of Facebook deactivation estimated using equation (1) for subgroups defined by the primary moderators in our pre-analysis plan. All variables are normalized so that the control group endline distribution has a standard deviation of 1. Error bars reflect 95 percent confidence intervals. See Section IC for variable definitions.

distinguishable for four indices, and opposite-signed for the final index. This provides some support for the view that effect sizes would not be systematically different.

(a) Figure 9: Heterogeneity, part 1

(b) Figure 9: Heterogeneity, part 2

6.8 Figure 10: Heterogeneous Treatment Effects for All Moderators + Table 5: Perceived Researcher Agenda

Read: Figure 10 searches across all moderators; Table 5 addresses experimenter-demand concerns.

Economic message. Effects appear to vary with social/psychological need and political/demographic characteristics, while demand-effect evidence is limited.

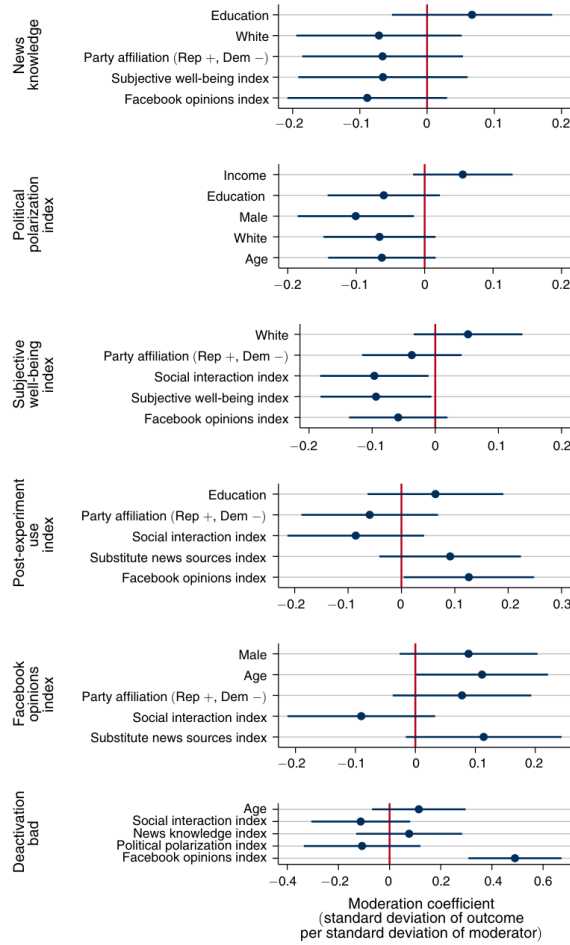


FIGURE 10. HETEROGENEOUS TREATMENT EFFECTS FOR ALL MODERATORS

Notes: This figure presents the moderators of local average treatment effects of Facebook deactivation estimated using equation (4). For each of the six outcomes, we present the five moderators with the largest moderation coefficients $\hat{\alpha}^k$. All outcome variables are normalized so that the Control group endline distribution has a standard deviation of 1.

(a) Figure 10: All Moderators

Variable	Treatment mean/SD (1)	Control mean/SD (2)	t-test p-value (1) – (2)
I don't think they had a particular agenda	0.43 (0.49)	0.44 (0.50)	0.59
Yes, wanted to show that Facebook is good for people	0.03 (0.18)	0.04 (0.19)	0.79
Yes, wanted to show that Facebook is bad for people	0.35 (0.48)	0.35 (0.48)	0.79
I am not sure	0.19 (0.39)	0.18 (0.38)	0.62
Observations	573	1,064	

Notes: The endline survey asked, "Do you think the researchers in this study had an agenda?" Columns 1 and 2 present the share of the Treatment and Control groups who gave each possible response. Column 3 presents p-values of tests of differences in means between the two groups.

(b) Table 5: Perceived Researcher Agenda

6.9 Figure 11: Distribution of WTA + Figure 12: Average Valuation

Read: Figure 11 shows dispersed WTA with a long right tail. Figure 12 shows treatment WTA falling after deactivation while control WTA rises.

Economic message. Facebook has large stated value, but deactivation reduces valuation, suggesting learning or projection-bias correction.

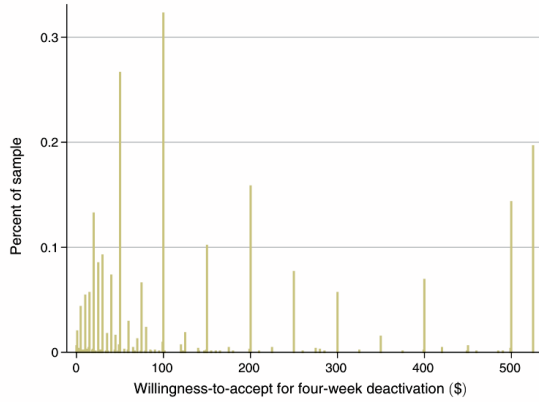


FIGURE 11. DISTRIBUTION OF WILLINGNESS-TO-ACCEPT TO DEACTIVATE FACEBOOK AFTER MIDLINE

This figure presents the distribution of willingness-to-accept to deactivate Facebook between midline and endline. All responses above \$525 are plotted at \$525.

(a) Figure 11: WTA Distribution

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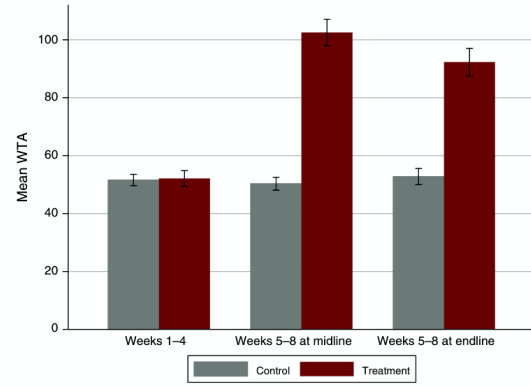


FIGURE 12. AVERAGE VALUATION OF FACEBOOK IN TREATMENT AND CONTROL

This figure presents the mean willingness-to-accept (WTA) to deactivate Facebook in Treatment and Control groups.

(b) Figure 12: Average Valuation

6.10 Table 6: Change in Facebook Valuation after Deactivation

Read: The coefficient on share deactivated is roughly -14 to -18 , about a 14% decline relative to treatment mean weeks 5–8 WTA at midline.

Economic message. The table is central to the welfare argument: experience with deactivation changed users' own valuation of future Facebook use.

TABLE 6—CHANGE IN FACEBOOK VALUATION AFTER DEACTIVATION

	(1)	(2)
Share of time deactivated	-14.36 (2.60)	-18.22 (7.73)
Observations	1,634	1,634
Winsorized maximum WTA	170	1,000
Treatment mean weeks 5–8 WTA at midline	103	135

Notes: This table presents estimates of equation (5). The dependent variable is the change in WTA for post-endline deactivation measured at endline versus midline. Standard errors are in parentheses.

Table 6: Change in Facebook Valuation after Deactivation

7 Short summary

- Deactivation reallocates time offline.
- It reduces news knowledge and political attention.
- It reduces some forms of political polarization.

- It increases subjective well-being.
- It lowers future Facebook use and future WTA.
- Facebook therefore creates large private value but also meaningful welfare and social costs.

8 Conclusion

8.1 Core conclusion

Facebook is valuable but not unambiguously welfare-improving. Users demand compensation to give it up, yet deactivation improves well-being, reduces polarization, and lowers future use.

8.2 Mechanisms

The mechanisms are time reallocation, changed news exposure, reduced polarizing exposure, and learning or habit disruption from experiencing deactivation.

8.3 Welfare interpretation

WTA implies large consumer surplus, but the decline in WTA after deactivation suggests baseline valuations may overstate fully informed welfare.

8.4 Policy takeaway

The lesson is not simply to eliminate social media. It is to measure platform welfare using both revealed valuations and experienced outcomes, while accounting for political and information externalities.